CNSCC.160 – Fundamentals of Communication Systems

Tutorial 1

Question 1:

What are the various transducers used in communication systems?

Transform physical message to electrical form

Question 2:

What do you understand by the term baseband signal?

Electric Equivalent of sound and computer

Question 3:

Briefly describe the source coding.

Remove the redundancy in message

Question 4:

Characterise the main areas of image processing: image compression, image enhancement and image understanding. State what their goals are and give examples.

图像效果增强 图像理解 减少冗余

Question 5:

Explain the difference between the high- and low-level image processing tasks.

输入都是图片, low→锐化,增亮

Question 6:

Explain the digital image representation.

Question 7:
Compare the principle of work of the human eye with the camera.

Question 8:
Explain the principle of work of the optical camera compared with the infrared (thermal) camera.

Question 9:
Why are the infrared (IR) rays able to penetrate given medium rather easily?